

Wits Quest (COMS3011A Project 6)

Master Architecture & Task Specification Guide

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Project: COMS3011A Software Design Project

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1. System Context & Software Background

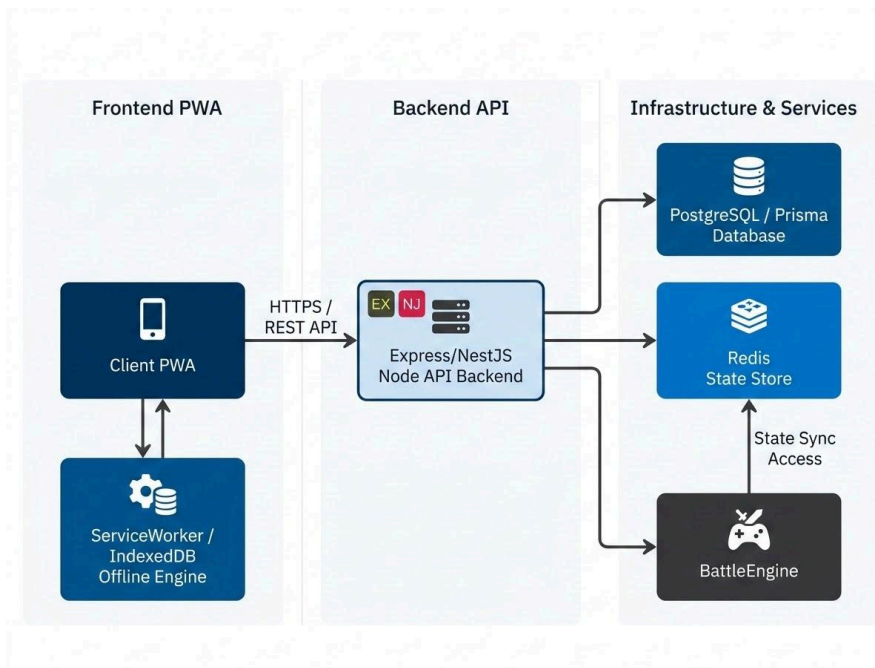
1.1 What is Wits Quest?

Wits Quest is a progressive, location-based web application (PWA) designed for the University of the Witwatersrand (Wits). Inspired by Pokémon GO and turn-based card strategy games, Wits Quest replaces traditional static campus tours with an interactive, location-gated trivia and collectible card battle experience.

The software combines several complex computer science domains into a single unified web platform:

1. **Geographic Information Systems (GIS) & Geolocation Verification:** Real-time interactive campus mapping with server-side mathematical validation of GPS coordinates.
2. **Offline-First Synchronous & Asynchronous Data Resilience:** Ability to explore campus and complete challenges in signal "dead-zones", caching attempts locally and securely syncing with the server upon reconnection.
3. **Turn-Based RPG Strategy & Rule Engine:** Card deck construction, stat attribute comparisons, CPU AI opponents, and real-time live WebSocket multiplayer.
4. **Game Economy & Progression Systems:** Card rarities, category attributes, daily streaks, achievement hooks, and atomic peer-to-peer trading.
5. **Security, Telemetry & Behavioral Anti-Cheat:** Speed-over-ground movement verification, anti-teleportation trajectory analysis, and trust-score computation.
6. **Authoring & Curation Governance Console:** Multi-stage content lifecycle management and campus foot-traffic heatmaps.

1.2 System Architecture Overview



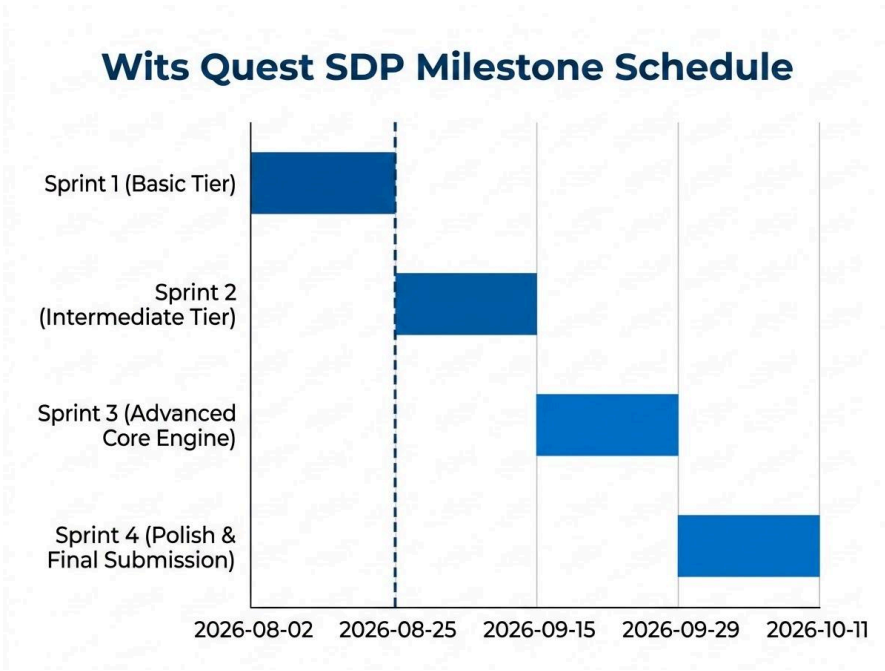
1.3 Recommended Technology Stack

- **Frontend Framework:** React (Vite) or Next.js with TypeScript.
- **Map & Spatial Rendering:** Leaflet.js / Mapbox GL JS with OpenStreetMap tiles.
- **Backend API:** Node.js with Express or NestJS (TypeScript).
- **Database & ORM:** PostgreSQL with Prisma ORM.
- **Real-Time Communication:** WebSockets (socket.io).
- **Offline Engine:** Service Worker API + IndexedDB.
- **Security & Auth:** Custom Authentication Engine (Wits Email validation @students.wits.ac.za).

2. Project Execution Strategy & Milestone Timeline

2.1 Strategic Rationale: The 2-Sprint Front-Load Push

To secure high marks early, the team will complete **100% of Basic Tier** and **100% of Intermediate Tier** by the end of **Sprint 2 (15 September)**.
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2.2 Official Milestone Schedule

Milestone Event	Deadline Date	Target Scope & Deliverable Focus
Group Selection	26 July 2026	Team formed (6 Members assigned)
Project Proposals	31 July 2026	Proposal submitted
Project Assignments	02 August 2026	Official project allocation
Milestone 1: Sprint 1	25 August 2026	Basic Tier Complete: GIS, Trivia, CPU Battle, Auth.
Milestone 2: Sprint 2	15 September 2026	Intermediate Tier Complete: Offline Sync, Async PvP.
Milestone 3: Sprint 3	29 September 2026	Advanced Core Engine: WebSocket Live PvP, 2PC Trading.

Milestone Event	Deadline Date	Target Scope & Deliverable Focus
Milestone 4: Final Submission	11 October 2026	Full System Polish: Territory Control, Elo Rating, Final Bug Fixes.

3. Analysis of Project Tier Expectations

3.1 Basic Tier Expectations

- **Campus Map & Events:** Leaflet/Mapbox map focused on Wits Campus showing active event markers.
- **Location Verification:** Server-side coordinate calculations before unlocking challenges.
- **Trivia & Marking Engine:** Server-side validation of trivia answers.
- **Card Collection & Deck Builder:** Card rewards per event; inventory and deck selection interfaces.
- **CPU Battle Minigame:** Turn-based stat comparison battle against a computer player.
- **Admin Console:** Interface for creating/placing events and defining card stats.

3.2 Intermediate Tier Expectations

- **Offline Signal Resilience:** Local attempt caching in ServiceWorker/IndexedDB.
- **Movement Anti-Spoofing:** Speed-over-ground verification (velocity calculations).
- **Asynchronous PvP Battle:** Async turn-based player challenges with persistent state.
- **Progression & Economy:** User profiles, XP/leveling, achievements, and daily streaks.
- **Guided Trails & Radar:** Sequential quest lines and unvisited spot compass.
- **Curation Governance:** Multi-stage content approval workflow (**Draft** -> **Published**).

3.3 Advanced Tier Expectations

- **Live WebSocket PvP:** Synchronous real-time turn battles and spectating.
 - **Heuristic / ML Anti-Cheat:** Behavioral profiling engine building player trust metrics.
 - **Algorithmic Dynamic Event Spawner:** Automated event placement along walkable campus paths.
 - **Ranked Matchmaking & Seasons:** Elo/Glicko-2 rating system.
 - **Territory & Zone Control:** Faction-based territorial control of campus zones.
 - **Atomic P2P Trading:** Secure 2-phase commit trading mechanism.
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4. Engineering & Software Task Matrix

4.1 SPRINT 1: Basic Tier Foundation (Aug 02 - Aug 25)

MEMBER 1 — Geolocation, Map & Spatial Engine

- **Task 1.1.1: Leaflet/Mapbox Spatial Initialization Engine**
 - *Technical Explanation:* Initialize engine restricted to Wits Campus bounding box. Implement `watchPosition` tracking.
- **Task 1.1.2: Server-Side Spatial Haversine Distance Engine**
 - *Technical Explanation:* Implement Haversine formula to compute distance in meters between user and target.
- **Task 1.1.3: Location Claim Verification API (POST `/api/events/:id/verify-location`)**
 - *Technical Explanation:* Verify `$distance ≤ event.radius` and `$time ∈ [activeStart, activeEnd]`.

MEMBER 2 — Turn-Based Battle Engine & CPU AI

- **Task 1.2.1: Turn-Based Battle Rules Engine (State Machine)**
 - *Technical Explanation:* Pure TypeScript state machine evaluating attribute choices and round winners.
- **Task 1.2.2: CPU Opponent Decision AI Engine**
 - *Technical Explanation:* Logic for `Easy` (random) and `Medium` (mathematically optimal) CPU moves.

MEMBER 3 — Database Architecture & Secure API

- **Task 1.3.1: Database Schema & Migration Architecture**
 - *Technical Explanation:* Design schema for `User`, `Event`, `Card`, `Match`, etc.
- **Task 1.3.2: Secure JWT Authentication Middleware**
 - *Technical Explanation:* Middleware inspecting `Authorization: Bearer <token>` headers.

MEMBER 4 — Admin Engine & Data Services

- **Task 1.4.1: RBAC Middleware & Security Guard**
 - *Technical Explanation:* Role-based route guard for `role: "ADMIN"`.
- **Task 1.4.2: Spatial Event Creation Service (POST `/api/admin/events`)**
 - *Technical Explanation:* API for creating new campus events with coordinates.

MEMBER 5 — Card Collection & Deck Construction Engine

- **Task 1.5.1: Inventory Query & Filtering Service (GET `/api/player/inventory`)**
 - *Technical Explanation:* Fetch cards owned by user with rarity filter parameters.
- **Task 1.5.3: Server-Side Deck Validation Engine**
 - *Technical Explanation:* Exactly 5 cards, max 1 `Legendary`, total stat cost ≤ 300 .

MEMBER 6 — Custom Auth Security & Audit Engine

- **Task 1.6.1: Custom Wits Student Email Auth Engine**
 - *Technical Explanation:* Validate `@students.wits.ac.za`. Hash passwords with `bcrypt`.
 - **Task 1.6.3: Rate-Limiting Security Middleware**
 - *Technical Explanation:* Restrict location/submission endpoints to max 10 requests/min.
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4.2 SPRINT 2: Intermediate Tier Completion (Aug 25 - Sep 15)

Member	Focus Area	Core Task Summary
M1	Anti-Spoofing	Velocity Engine: Detect impossible speeds ($> 5.0 \text{ m/s}$).
M2	Async PvP	State Store: Persistent turn history for player challenges.
M3	Offline Sync	IndexedDB Queue: Cache event attempts locally when offline.
M4	Curation	Workflow Engine: Manage DRAFT $\rightarrow$ PUBLISHED states.
M5	Progression	XP/Leveling Engine: $XP = 100 \times \text{level}^{1.5}$ math integration.
M6	Matchmaking	Level-Proximity Query: Find opponents within ± 3 levels.

4.3 SPRINT 3: Advanced Core & Live Engines (Sep 15 - Sep 29)

[ALERT] Advanced Implementation Notice: Sprint 3 introduces real-time state synchronization via WebSockets. All state transitions must be verified by the Redis cache to ensure session consistency.

- **Member 1 (Spawner):** Algorithmic placement of events along GeoJSON path grids.
 - **Member 2 (Live PvP):** `socket.io` implementation with 30s turn countdown clocks.
 - **Member 3 (Trading):** Two-Phase Commit (2PC) engine for atomic card ownership swaps.
 - **Member 4 (Heatmaps):** Spatial aggregation service processing coordinate clusters into intensity matrices.
 - **Member 5 (Ranked):** Elo/Glicko-2 rating recalculation formula: $R_{\text{new}} = R_{\text{old}} + K(S - E)$.
 - **Member 6 (Trust Engine):** Composite Trust Score (\$0-100\$) based on behavioral heuristics.
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4.4 SPRINT 4: System Polish & Submission (Sep 29 - Oct 11)

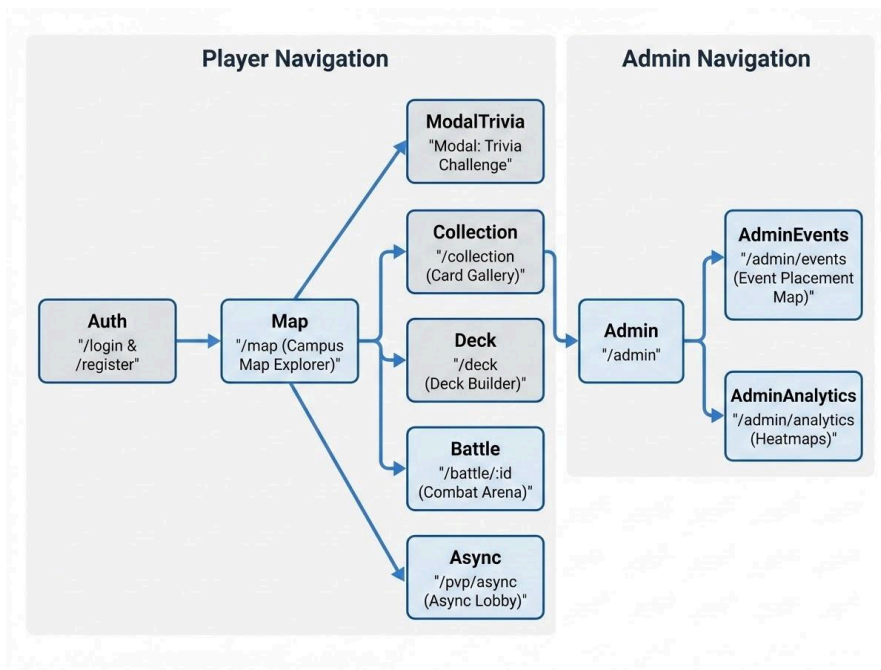
- **Spatial Polish:** PostGIS indexing optimization ($< 15\text{ms}$ query times).
 - **Battle Balancing:** Monte-Carlo simulation (10,000 matches) to verify attribute balance.
 - **Transaction Audit:** Security verification of P2P trade race conditions.
 - **Territory Control:** Final engine for dividing campus into 5 faction zones.
 - **Dockerization:** Production `docker-compose` orchestration for final hand-off.
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5. Definition of Done & QA Standard

Note on Quality: For any software task to be marked as complete, it must satisfy the following checklist:

1. **Code Review:** Merged via Pull Request with at least 1 peer approval.
2. **Automated Testing:** Unit or integration tests written and passing.
3. **API Verification:** Verified with valid status codes (200, 201) and JSON payloads.
4. **No High Blockers:** Zero unresolved high-severity bugs on the task.

6. UI/UX Screen Specifications



6.1 Selected Screen Components

- **Screen 2: Campus Map:** Full-screen Leaflet map, pulsing blue GPS dot, and floating XP/Level progress badge.
 - **Screen 4: Card Gallery:** Responsive grid with rarity-specific borders (Gold, Purple, Blue, Gray) and "Scrap for Essence" buttons.
 - **Screen 6: Combat Arena:** 30-second circular countdown clock, card-vs-card collision animations, and Elo change indicators.
 - **Screen 10: P2P Trade:** Split-screen "Your Offer" vs "Their Offer" boxes with an 8% stat-difference validation badge.
 - **Screen 15: Admin Heatmaps:** Glowing intensity overlays of campus movement with time-range filters.
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